

SESHADRIRAO GUDLAVALLERU ENGINEERING COLLEGE

(An Autonomous Institute with Permanent Affiliation to JNTUK,Kakinada)

Seshadri Rao Knowledge Village, Gudlavalleru – 521 356, Krishna District

Department of Electronics and Communication Engineering

Registration for B.Tech MainProject

AcademicYear:2025-2026

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Batch No: C5

Name of the Students:

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Broad Area of Proposed Project Work

Embedded Systems, Assistive Technology, IoT-based Smart Home Automation.

Title of the Proposed Project Work

Braille-Based IoT Home Automation for Multi-Disability Empowerment.

Abstract

Visually impaired and speech-disabled individuals face significant challenges when using modern home appliances. Most smart home solutions depend on mobile apps or voice commands, which are not accessible to them. To tackle this issue, the proposed project introduces a Braille-Based IoT Home Automation System. This system allows blind and speech-disabled users to control household devices independently. It combines a 6-dot Braille input keypad with an ESP32-WROOM-32 IoT controller. Users can input tactile commands that correspond to specific appliances, such as lights, fans, and door locks. Each Braille code is interpreted by the ESP32-WROOM-32 and sent over Wi-Fi to manage the devices. To enhance usability, the system will have two types of feedback. Users will receive auditory feedback through distinct beep codes, while caretakers will get visual feedback from an LCD/LED display. The system will also include emergency features that allow predefined Braille inputs to trigger alarms or send SMS alerts. The proposed solution aims to be low-cost, inclusive, and safe. It will provide a flexible platform for smart home automation. The novelty of this project lies in combining Braille-based tactile interaction with IoT automation, helping to close the accessibility gap in smart homes for users who are visually and speech-impaired.

Implementation Tools Required

Hardware

- ESP32-WROOM-32 microcontroller (e.g., Espressif ESP32-WROOM-32-WROOM-1)
- Custom Braille keypad (6-dot tactile interface)
- Touch sensors (TTP223 Digital Capacitive Touch Sensor Module)
- Relay modules (e.g., 5V 8-Channel Relay Module)
- Buzzer (e.g., active buzzer module)
- LCD display (e.g., 16x2 LCD with I2C)
- Power supply unit (e.g., 5V DC adapter with Li-ion battery backup)
- Wi-Fi router
- Emergency GSM/SMS module (e.g., SIM800L)

Software

- Arduino IDE
- Programming languages: C++

References

- ICMAR 2023 – “Raspberry Pi based Braille keyboard design with audio output.”
URL: <https://as-proceeding.com/index.php/icmar/article/download/1306/1245/2406>
- IEEE ICCA 2024 – “Scalable Smart Home Management with ESP32-S3.”
URL: https://www.researchgate.net/publication/390217577_Scalable_Smart_Home_Management_with_ESP32-S3_A_Low-Cost_Solution_for_Accessible_Home_Automation
- T. V. Lakshmanan, “Assistive IoT Technologies for the Differently-Abled,” IEEE Access, 2022.
URL: https://www.researchgate.net/publication/312001963_The_Internet_of_Things_and_Assistive_Technologies_for_People_with_Disabilities
- AP/STA communication between esp32s.
URL: “https://youtu.be/2z7Ow4PjJRc?si=FIW6IwR_eVhzwoQw”
- ESP-now communication protocol for esp32.
URL: “<https://youtu.be/xOLG-88Ld3A?si=BnvgO9c1dykmINd2>”

Signature of the students

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Remarks by the Guide:

Signature & Name of the guide with date

(Mr.V.Vittal Reddy)

Signature of B.Tech with Project Coordinator date

Signature of HoD with date

(Dr.B.Raja Sekhar)